Gears, motor and first power

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare a gear-based prediction with a supervised first powered test.

SUCCESS CRITERIA - I can...

1. Check gear mesh and secure connections before power.
2. Calculate ideal wheel speed from measured or supplied rpm.
3. Record the first test results and one change to the design.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “circumference” and “gear ratio” to explain your answer. Sentence starters are optional.

circumference: distance around the edge of a circle

gear ratio: a comparison of gear tooth counts or rotation rates

assumption: a condition accepted for using a model

01 Warm-up: bring back an earlier idea

A **30**-tooth driver turns a **150**-tooth gear. **Give ideal output**/input torque and speed ratios and say what you treated as true about energy losses.

Word help: torque = the turning effect of a force about an axis

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

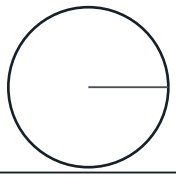
02 Worked example: explain the prediction

Reference: rotation rate = turns/Δt
v = circumference × rotation rate.
Turns per second, no slip. Get first-power approval; secure car and keep loose items clear. Disconnect before adjusting gears.

Worked reasoning: Supplied wheel tests show **4** rotations in **2 s**. With circumference **0.20 m** and no slip, predicted speed is (**4/2**)×**0.20**=**0.40 m/s**. Ground speed may differ if wheels slip.

Your turn: Draw one wheel rotation and the predicted travel distance. **Identify why you treat the** wheels as rolling without slipping.

Sentence starter: I predict ... . I used ... to make this prediction because ... .



Radius / diameter: ____

Turns per second: ____

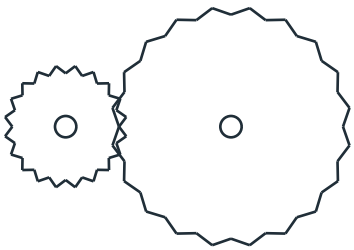
Travel per turn: ____ Ground speed: ____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Sketch your driver and driven gears. **Label tooth counts** and **name the driven axle**. Note any rubbing or loose mesh before power.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Driver Driven

Marzano check - circle: ① ② ③

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Continue down each column.

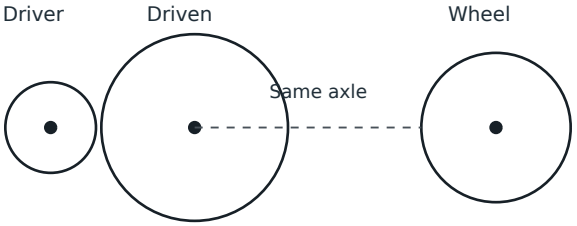
Sentence starters are optional. You may plan in any language.

04 Apply the physics

Under load, a small classroom car motor turns at **480 rpm**. A **12**-tooth driver turns a **36**-tooth driven gear, and wheel diameter is **0.050 m**. **Label the drivetrain**, predict ideal ground speed with no slip and **calculate time over 3.0 m**.

Word help: ideal = using stated simplifying conditions

Sentence starter: I predict … . I used … to make this prediction because … .



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3

05 Explain your reasoning

The actual car is slower than the prediction. **Name two things your prediction** treats as true that you should check and one measurement that could separate their effects.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A bar magnet is cut into **4** pieces. Does this produce isolated *north* and *south* poles? **Sketch the usual result**. Use field arrows and spacing to **support your explanation**.

Sentence starter: My design includes … . It does what is needed by … .

N	S

Marzano check - circle: 1 2 3

07 Exit check

A no-load motor rate is **600 rpm** and loaded rate is **450 rpm**. Compute the relative reduction and **explain why no-load rpm overpredicts** travel speed.

Sentence starter: I predict … . I used … to make this prediction because … .

Checkpoint / measurement	Evidence	Next action
___	___	___

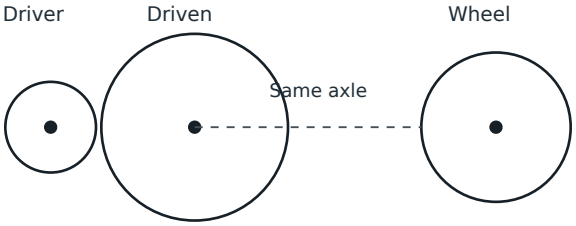
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A wheel of radius **0.025 m** rotates at **240 rpm**. **Find ideal ground speed**. Measured speed is **0.50 m/s**; **find the ratio measured**/ideal and avoid treating that ratio alone as proof of slip.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3